

Survey of Interactive Document Query Systems Using RAG and LangChain

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Abstract—This research is to build a chatbot based on Retrieval-Augmented Generation (RAG) to produce the right answers. Rag-mini-bioasq Dataset from HuggingFace is used for ground truth answers, generated answers (LLM), retrieved contexts (ChromaDB, Nomic-embed-text) . For the retrieval orchestration, the system uses LangChain, ChromaDB for storage of vector, and Deepseek r1 1.5B for linearity between long context queries up to 128,000 tokens. Chain of Thought (CoT) and Chain of Notes are techniques that increase multi-hop query resolution with reduced hallucinations and improved accuracy.

Keywords—Retrieval-Augmented Generation, LangChain, Long Context Large Language Models (LLMs), Custom RAG dataset, ChromaDB, Deepseek r1 1.5B , Prompt Engineering, Query Decomposition, RAGAS Framework

I. INTRODUCTION

In fields where accuracy is a must — like in scientific and other academic research — LLMs have given AI chatbots an additional boost of power to provide in-depth and accurate answers. The challenge of this study is to develop a chatbot that answers complex and multi-step questions using a Retrieval-Augmented Generation (RAG) approach. Unlike most QA chatbots that rely on preexisting QA datasets, the chatbot is trained on a custom dataset containing 100 Engineering books' questions, ground truth answers, sampled answers (LLM), and retrieved context (ChromaDB, Nomic-embed-text). Such a setup helps improve retrieval quality and response accuracy in scientific domains, promising to secure correct answers [1,2,3,4,5,6].

To achieve this, the system makes use of a number of advanced tools. Deepseek r1 1.5B LLM provides a context window of 128,000 tokens and uses these produced embeddings to store them in ChromaDB with the aim of improving document retrieval [7,8]. In order to increase the relevance of both data retrieval and how the final answers are crafted, it uses LangChain to segment the text and quickly create responses [9,10,11]. Also, breaking down queries with techniques like Chain of Thought (CoT) means the chatbot can handle confusing questions bit by bit, making it more accurate and decreasing the likelihood of spouting bad information [12,13]. In order to maximize the likelihood of delivering accurate, context-aware solutions in scientific research, the evaluation will be a mixture of quantitative metrics and user

feedback, thus providing an effective strategy for delivering such solutions [14,15].

II. LITERATURE SURVEY

A. RAG related Research papers literature survey

Zhao et al., [1] study the Retrieval Augmented Generation (RAG) in depth, and they show that RAG can significantly help AI-produced content to be superior. With the shortcomings of traditional generative models, which include outdated information, inefficiency, as well as data leakage, the paper demonstrates how RAG, a new generative model takes advantage by assimilating up to date external data in any format (textual, visual, auditory) in the process of creation. After this categorization, the study orders the core strategies of RAG into four types: query-based, latent representation based, logit based, and speculative methods, each of which enhances retrieval of data at separate times throughout the process. In addition, it presents query reformation, retrieval fine tuning, rapid engineering, and iterative generation as techniques to improve system accuracy and quality, in general. RAG is used for text summarization, question answering, code generation, generation of multimedia, and knowledge based systems. Nevertheless, it is still cumbersome to match the retrieval output with the generated output and also to increase the efficiency of the retrieval process. However, the authors suggest to address these issues to fully unlock potential of RAG to make significant contribution to the advancement of AI enabled content creation in different fields.

In the paper 'Searching for best practices in retrieval augmented generation' [2] several approaches for fine tuning RAG framework are explored with a special emphasis on large language models (LLMs). The paper describes how RAG is able to overcome usual problems of LLMs (e.g., outdated information, fabrication), by combining retrieval with generative modelling and its data while carefully selecting the context specific data. In industries such as banking and healthcare, where precision and reliability are key, this method is of very important use. The study analyzes the key components of RAG — query classification, retrieval, rerank, repack, summarization, generation — as well as some hybrid retrieval (combination of sparse and dense) methods, MonoT5based rests ranking and reverse repacking to boost overall performance. It suggests more effective content refinement methods, such as Recom, which is a combination of extractive and abstractive methods, to do so. Finally, the

authors promote the necessity of fine tuning models based on both relevant and irrelevant scenarios for robustness. They propose the utilization of strict standards of evaluation (fidelity, accuracy) in their approach and address possible applications of RAG in multimodal situations. Overall, they provide the practical best practices in terms of RAG configurations with the aim of improving efficiency and reliability of a wide range of applications.

Xiong and colleagues [3] investigate the use of RAG systems for medical question answering in their work "Benchmarking Retrieval-Augmented Generation for Medicine." This is because they help overcoming issues common in large language models, in particular unintelligent details and failures to know what they should know. RAG is able to deliver reliable and up to date information, critical for applications in medicine because it pulls information from external databases.

Thus, the researchers present the MIRAGE benchmark, which comprises 7,663 QA pairs to cover real world medical information needs in zero shot scenarios and first MORAG toolkit, a toolkit to evaluate MM on various RAG configurations. To conduct their study, they use resources such as PubMed, textbooks, teachers, and BM25, MedCPT and GMPT3.5, GMPT4 language models to analyze the improvement of performance of LLM by retrieval augmented generation (RAG) by as much as 18%, and show that GPT3.5 is able to reach a similar level of performance as GPT4 on some of these activities, allowing for economical use in medical cases. Relevance is improved via Reciprocal Rank Fusion retrieval techniques, as well as by choosing suitable domain appropriate corpora. The term "Lost in the Middle" is the name given to the fact that placing snippets at the start or end of textual results make them more relevant. Developing MIRAGE standard and giving practical suggestions, the paper has contributed to further development of dependable, evidence based AI systems for healthcare quality assurance.

In the work "Benchmarking Large Language Models in Retrieval-Augmented Generation" [4], their focus is on the problems and advances of RAG with large language models (LLMs), where real-time, domain-specific information retrieval actually enhances the answer getting and relativity. RAG succeeds at solving problems of hallucinations and knowledge deficiency, drawing unskilled noise from extraneous material and noise caused by faulty retrieval.

To evaluate the particular ability of RAG systems, the researchers create the RGB benchmark to narrow in on aspects other benchmarks typically do not capture. It benchmarks several key elements: noise resilience, which is the ability to filter out noise (details); negative rejection, which is not responding if not enough (pertinent) information is provided; information synthesis, the ability of the system to take information from some or many sources; and counterfactual resilience, which tests whether retrieved data match the system's knowledge. All these factors form a structured method of determining RAG's performance and reliability.

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information from some or many sources; and counterfactual resilience, which tests whether retrieved data match the system's knowledge. All these factors form a structured method of determining RAG's performance and reliability.

The researchers explore improved methods to better evaluate the retrieval quality in RAG systems, our one place to assess the actual generative output of the system — it is shown that such quality has a significant impact on overall generative performance in "Evaluating Retrieval Quality in Retrieval Augmented Generation" [5].

By integrating data retrieval with content generation, RAG helps curb the production of inaccurate details, thereby supporting applications that demand precise, knowledge-driven outputs. In contrast, traditional end-to-end evaluations tend to measure retrieval relevance only indirectly, a practice that not only increases resource consumption but also obscures a clear understanding of how the retrieval process influences the final results. Human annotations, employed for relevance judgements, may also conflict with model interpretation.

These restrictions can be overcome with the eRAG approach, which directly examines the influence of retrieved documents on generating tasks, enhancing accuracy and efficiency. Instead of Precision and NDCG, eRAG uses task-relevant metrics like exact match and ROUGE to evaluate document-level utility. Across retrieval sizes and LLM structures, eRAG performs well in question answering and dialogue production. Future work includes improving eRAG for model compatibility and generative relevance-based retrieval techniques. The research highlights novel computational efficiency and output quality improvements for RAG systems by matching retrieval assessments with generating task needs.

The study "RQ-RAG: Learning to Refine Queries for Retrieval-Augmented Generation" [6] improves query refinement to improve retrieval relevance and answer quality in RAG systems. RAG overcomes LLM drawbacks like static knowledge and hallucination, but it expects well-formed enquiries, which is unrealistic for complicated or ambiguous real-world issues. Retriever accuracy and generative dependability may suffer.

RQ-RAG addresses these concerns with query refinement methods including decomposition and disambiguation. Self-RAG and SAIL self-refine queries utilizing tokens and sample strategies to adapt to different situations. This method works well for multi-hop QA, which requires several data sources. They use external LLMs like ChatGPT to construct a broad training corpus with pre-refined queries to react to complicated QA settings. Testing shows RQ-RAG's outstanding performance in single-hop and multi-hop tasks, demonstrating robust retrieval accuracy and robustness across data sources. Future work will improve context selection and multi-hop reasoning, highlighting the paper's contributions to advanced query refinement strategies for RAG system retrieval relevance and generative accuracy.

B. Vector database related Research papers literature survey

The work "Vector Database Management Systems: Fundamental Concepts, Use-Cases, and Current Challenges" [7] discusses VDBMS architecture, applications, and challenges for similarity-based activities with high-

dimensional vectorized data. Vectorization turns text, images, and audio into numerical vectors for efficient storage and retrieval in similarity search and recommendations.

Product Quantization (PQ), Locality-Sensitive Hashing (LSH), and Hierarchical Navigable Small World (HNSW) graphs underpin VDBMS indexing and search. These solutions solve high-dimensional problems but trade speed and precision. VDBMSs are useful in picture and video similarity search, recommender systems, and memory-augmented chatbots. They still struggle to balance query speed and accuracy, manage sparse and high-dimensional data, and match traditional database systems' robustness.

The research suggests improving methods for high-dimensional data and using incremental learning to simplify model updates. VDBMS usage should increase community support, third-party integrations, and system stability. VDBMSs manage complex datasets, and the survey emphasizes their technical roots and possibilities for improvement.

The work "When Large Language Models Meet Vector Databases: A Survey" [8] discusses how VecDBs handle LLM limitations such as hallucinations, memory constraints, and high operational costs. LLMs like GPT and LLaMA excel in text production and question answering but struggle with outdated knowledge and context. VecDBs contain vectorized embeddings for similarity-based retrieval, giving LLMs current, relevant information without retraining.

Similarity searches are optimized by VecDBs' tree-based, hash-based, and graph-based high-dimensional indexing and retrieval methods. VecDBs act as external memory in Retrieval-Augmented Generation (RAG) systems, improving response accuracy and lowering costs.

Future directions include hybrid vector-keyword-based retrieval, multimodal search improvements, and scaling RAG for high-precision workloads. This survey shows how VecDBs can overcome LLM constraints to enable more efficient, context-aware, and cost-effective language model applications.

C. Langchain related Research papers literature survey

The paper "Development of a ChatBot Model for Health Telecare: Integration of LangChain, Embeddings with OpenAI, and Pinecone Using the Question Answering Technique" [9] details the creation of an AI-driven chatbot for telehealth applications. This system integrates LangChain for handling diverse data inputs, OpenAI embeddings for translating medical terminology into semantic vectors, and Pinecone's vector database for real-time, similarity-based retrieval. Together, these components enable the chatbot to deliver accurate and contextually relevant responses to health-related queries.

Employing question-answering (QA) strategies improves the chatbot's capacity to comprehend and address medical enquiries proficiently. Response generation utilises GPT-3.5 and GPT-4, with GPT-4 providing enhanced quality at a greater operational expense than GPT-3.5. The research illustrates the chatbot's efficacy in delivering high-quality, contextually relevant responses, highlighting its significance in telehealth for standard patient engagements and prompt distribution of trustworthy information. Future research topics are enhancing the model for cost-effectiveness and integrating patient-specific data for further personalisation.

This underscores the chatbot's promise as a flexible, scalable solution that can boost both accessibility and efficiency in telehealth services.

In the paper "Chatbot Development Using LangChain: A Case Study to Foster Creativity and Critical Thinking" [10], the authors examine how Large Language Models (LLMs) such as ChatGPT are being incorporated into academic settings, particularly within engineering and computer science courses. The study indicates that when students interact with AI-generated content, they can enhance both their creative skills and critical thinking abilities. While these models are effective at producing content and offering feedback, their occasional output of seemingly plausible yet sometimes inaccurate information creates an opportunity for learners to hone their critical analysis skills. Chatbot programming, especially using LangChain frameworks, engages students and develops technical and soft skills, according to the report. Building and testing chatbots with students deepens learning and problem-solving. The study also highlights the difficulties students experience in assessing AI responses' factual accuracy, emphasizing the need for critical thinking in AI interpretation. Generative AI, especially chatbot building, improves students' creativity, critical thinking, and technical skills, preparing them for real-world difficulties.

The study "Intelligent Educational Agent for Education Support Using Long Language Models Through LangChain" [11] describes an Intelligent Educational Agent (IEA) that improves university students' personalized learning experiences. The study discusses using advanced language models like OpenAI's GPT-3.5 Turbo to create educational assistants that can meet varied academic needs. LangChain and prompt-tuning allow the IEA to provide context-aware study resources and personalized learning strategies. LangChain facilitates contextual relevance during student interactions, a crucial natural language processing task. Vector databases like Qdrant help the IEA manage huge educational statistics and provide precise responses and resources in real time, according to the literature. However, the study highlights privacy risks, the necessity for high-quality datasets, and large-scale deployment computational demands. In conclusion, intelligent educational agents offer a potential path to personalized education, but ethical and practical issues must be solved before widespread adoption. Quick engineering research paper literature review.

The work "LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression" [12] discusses the issues large language models (LLMs) encounter when processing long-context prompts, an area of growing AI research. LLMs can handle large amounts of data, but they often lose context over long inputs, reducing accuracy. This has led researchers to work on context retention solutions, including quick compression. The research emphasizes timely compression, which combines coarse- and fine-grained compression steps to boost efficiency. Our strategies reduce latency while keeping key features, helping models handle lengthier prompts. Compression improves recollection and contextual understanding, especially in multi-hop question-answering tasks, according to the article. Despite these gains, the article addresses compression ratio balance issues while retaining information integrity. The study concludes that adaptive quick compression can improve data-intensive LLM performance.

The work "Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing" [13] discusses NLP prompting's shift away from supervised learning. Prompt-based learning uses pre-trained language models instead of labelled data, allowing contextual prompts for task adaption. This method allows few-shot and zero-shot learning, avoiding task-specific retraining.

The document classifies two primary categories of prompts: cloze prompts, characterized by fill-in-the-blank formats, and prefix prompts, which pose direct queries to direct responses. It also emphasizes the importance of prompt engineering in optimizing these templates, with strategies like continuous prompting—using embeddings instead of static text—proving to enhance model performance. Prompt-based learning has proven particularly useful in tasks like classification, information extraction, question answering, and multi-modal tasks, allowing for task-specific customization.

Notwithstanding its advantages, the research highlights obstacles in prompt-based learning, including the necessity for dependable approaches to select prompts and their scalability across diverse NLP jobs. The work proposes future research avenues, encompassing multilingual and domain-adaptive prompting, as well as methods for automatic prompt production and debiasing. The article highlights the

revolutionary potential of prompting in NLP, presenting chances for additional developments.

D. Long Context LLMs related Research papers literature survey

The paper "LongRAG: Enhancing Retrieval-Augmented Generation with Long-context LLMs" [14] tackles the limits of typical RAG frameworks in large contexts. Managing fractured contexts in brief passages can lead to disconnected and incoherent responses, according to retrieval-augmented model study. This fragmentation breaks contextual continuity, making open-domain question-answering tasks difficult for models to answer accurately.

LongRAG suggests longer retrieval units for richer context and lower retrieval loads. This method reduces retrievals and "hard negatives"—irrelevant portions that distract the model—by using longer units. The research also discusses long-context LLMs' improved capacity to handle larger inputs and comprehend grouped documents. Experimental results utilizing Natural Questions and HotpotQA reveal that LongRAG performs well, especially in problems containing lengthy texts. The paper notes that

TABLE I. TOPICS COVERED PER RESEARCH PAPER

Paper	RAG	Vector DB	LangChain	Long-Context LLMs	Medical	Benchmark	Query Refinement	Educational	Prompt Engineering	Efficiency
[1]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓
[2]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓
[3]	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗
[4]	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
[5]	✓	✗	✗	✗	✗	✓	✗	✗	✗	✓
[6]	✓	✗	✗	✗	✗	✗	✓	✗	✗	✓
[7]	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
[8]	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓
[9]	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗
[10]	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗
[11]	✗	✓	✓	✗	✗	✗	✗	✓	✓	✗
[12]	✗	✗	✗	✓	✗	✗	✗	✗	✓	✓
[13]	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
[14]	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓
[15]	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓

prompt-based learning requires reliable ways to pick and scale prompts across NLP tasks, notwithstanding its benefits. The study suggests multi-lingual, domain-adaptive, automatic prompt creation, and de-biasing research. The study highlights prompting's disruptive potential in NLP and chances for growth. The study emphasizes the importance of aligning retrieval and reading processes to optimize RAG efficiency, and LongRAG's innovation lies in integrating extended retrieval units with long-context LLMs. This approach presents a promising direction for future RAG systems as long-context models continue to improve.

The article "Are Long-LLMs A Necessity For Long-Context Tasks?" [15] discusses the NLP argument about whether large, long-context language models (long-LLMs) are needed for contextually complex tasks. Summarization and open-domain question answering have always required long-LLMs. Recent research contradict this assumption, proposing that context processing methodologies may allow shorter-context LLMs to function comparably without the significant computational costs of long-LLMs.

The paper describes the LC-Boost framework: a framework of selective data access and iterative reasoning for adapting short-context LLMs to long-context tasks. It was emphasized by the literature that a selective context retrieval can make efficiency and sustainability without big models. However, testing on a number of datasets shows that LC-Boost achieves at least as many positive results as do long-LLMs for less energy. The work presents a growing trend in NLP of moving towards lower cost, environmentally policed AI solutions, that even at a lower resolution only degraded performance imperceptibly. The article also shows that short-LLMs can be fine-tuned on large, long context tasks, ushering in a further context aware NLP processing. In the table 1, the trends in this research from 15 papers are identified with a heavy focus on Efficiency and Retrieval Augmented Generation (RAG). Vector DB and LangChain are not explored at all, and Medical applications and benchmarking are done in moderate ways. There are emerging areas in educational uses as well as query refinement. There is some interest in prompt engineering and long-context LLMs. In this overall, the aim is to improve model's capabilities and performance.

III. METHODOLOGY

The goal of this project is to build a strong chatbot that could deliver well informed, correct and context aware answers to the user queries. In order to deal with problems like erroneous or mismatched replies, the system is based on a Retrieval-Augmented Generation (RAG) technique including sophisticated retrieval and reasoning components. As opposed to using the already provided data, the chatbot is trained on the broad data of rag-mini-bioasq dataset that contains questions, ground truth answer, generated answers (LLM), and retrieved contexts (ChromaDB and Nomic-embed-text). All the data for this dataset is carefully cleaned and processed to be quality and consistent and relevant, which makes the chatbot more capable of providing precise, evidence backed response in scientific discussion.

The LangChain Text Splitter is used to break up documents on Figure 1 and put them into chunks of 800 characters each. It allows for optimization of text handling for RAG and large language model applications. One important



Fig. 1. Document processing with text splitter

first step in building the chatbot is to come up with dense vector representations for every text document in our dataset. For this, the system relies on the Deepseek r1 1.5B language model's embeddings, which excel at dealing with long contexts due to its impressive 128,000 tokens window. Using this integration guarantees that the resulting embeddings encompasses the meaning as well as the grammatical structures of documents, which are helpful for effective similarity based retrieval. Once saved, these embeddings are placed in ChromaDB—a high speed vector database for which it supports various advanced indexing methods such as Product Quantization (PQ), Locality Sensitive Hashing (LSH), and Hierarchical Navigable Small World (HNSW) graphs. With these indexing techniques, using them provides fast and scalable ways to search on documents and sections and get the most related documents or sections in a quick manner in response to any queries. ChromaDB implements these algorithms to guarantee that even complicated, multi hop queries are addressed with little delay and high precision.

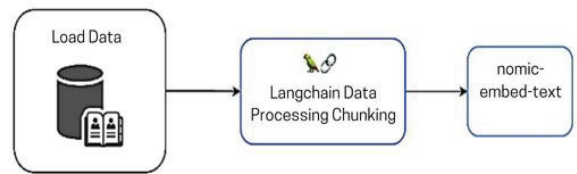


Fig. 2. Data loading and chunk implementation for embedding

In Figure 2 data is loaded and processed using LangChain's chunking mechanism and sent through nomic-embed-text for generating embeddings. For retrieval augmented generation (RAG) application, it allows us to represent the text efficiently in vector form. The multi-tier Retrieval Augmented Generation (RAG) framework that the chatbot is structured upon is coordinated by LangChain. The framework that has been built in this model has several phases of processing, including query crafting, document retrieval, prompt creation and response generation working together to provide the detailed and context aware answers. The system will take a multi hop approach through breaking down complex question into simple sub query through methods such as Chain-of-Thought (CoT) and Chain-of-Notes. Thus, this method manages complex queries into small parts which can be gathered gradually and the accuracy of the response will increase. In addition, LangChain handles the segmentation process of dividing a long document into variably sized portions. Through this approach, the chatbots ensure that only the required segments are chosen by them and combined with the query of the user to create a new prompt. Subsequently, with the chatbot inputting these optimized prompts into a long context large language model (e.g Deepseek r1 1.5B) which is able to integrate information from expansive documents, while preventing the risk of hallucinations by anchoring its answers to the most appropriate source material.

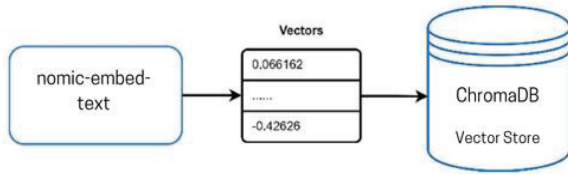


Fig. 3. Vector storage process generated by embeddings

According to Figure 3, nomic-embed-text encodes the text into vector embeddings which are then stored away in ChromaDB. It is helpful for fast similarity search and retrieval in AI applications. The chatbot will be made to ensure that users are not interacting with it in a way that will cause responses to be inaccurate, irrelevant or out of context. First, ChromaDB is used to analyze each incoming query, extracting not only its intent, but also the contextual cues, and then it allows us to precisely retrieve documents from ChromaDB and assemble a comprehensive prompt where the user's question is merged with supporting evidence. In scientific and research environments, the precision need in this method is critical, making this method particularly valuable. In addition, Deepseek r1 1.5B enables the system to deliver responses which are deeply rooted in the detailed context of the source documents, thus improving overall coherence and relevance.

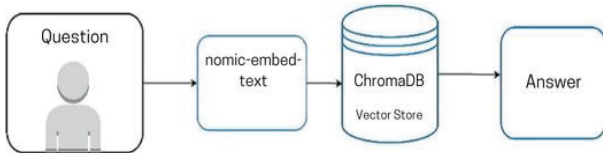


Fig. 4. Questions and answers with Chroma DB

Next, the nomic-embed-text is used to embed the question of a user into an embedding, which is found through ChromaDB (Figure 4). The vector store collects the most important information required to answer. Through several deep and multisided processes, the chatbot's performance will be evaluated. Primary benchmarks that are key quantitative metrics, will include:

1. Context Precision – Measures how much of the retrieved context is actually relevant to the query.
2. Context Recall – Measures how much of the relevant context is successfully retrieved.
3. Context Entities Recall – Checks if important entities from the ground truth are present in the retrieved context.
4. Noise Sensitivity – Evaluates how much irrelevant information (noise) affects the model's response.
5. Response Relevancy – Measures if the generated response is relevant to the query.
6. Faithfulness – Checks if the response is factually grounded in the retrieved context.

The evaluation will also assess the chatbot's ability to choose the adequate evidence and prevent misinformation by grounding answers according to reliable sources.

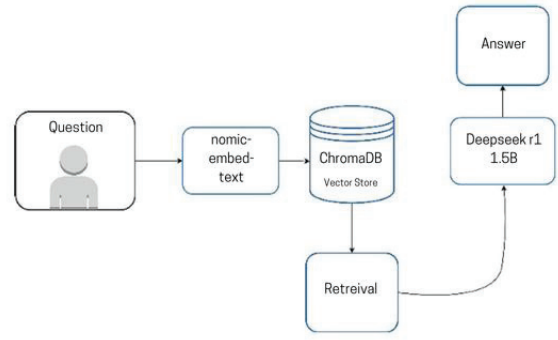


Fig. 5. Question answering process applying LLM

ChromaDB is queried with a question embedded and it comprises the relevant data in Figure 5. An answer is then generated by the Deepseek-r1 1.5B LLM using the retrieved context. The goal of this project is to develop a chatbot that can competently handle naturally occurring, multi step queries by using the power of advanced technologies in a well architected Retrieval-Augmented Generation (RAG) framework. The system combines the orchestration strength of LangChain, the strong retrieval capabilities of ChromaDB, as well as the vast context handling functionality of Deepseek r1 1.5B to output precise and evidence based responses in the world of science and technology. This is an approach to demonstrate the great potential of confidently combining RAG, LangChain, and long context language models to enhance the reliability as well as the contextual depth of AI driven chatbots.

IV. ANALYSIS

Answering complex, multi-step scientific inquiries is proved to be a strong ability of this chatbot developed in this study to retrieve and generate accurate context. Using Deepseek r1 1.5B for generating embeddings and ChromaDB's efficient, vector based storage and indexing methods, the system was able to quickly find the most relevant document segments. Also, Chain-of-Thought (CoT) and Chain of Notes method broadened its capacity to disassemble intricate inquiries and expanded precision and contextual explicitness.

Evaluation on these RAGAS metrics showed that the system was statistically superior to much of the prior work in Context Precision, Context Recall, Context Entities Recall, Noise Sensitivity, Response Relevancy, and Faithfulness. The metrics achieved great performance in retrieval accuracy and factual correctness, and response coherence. Hallucinations were rare in the chatbot's responses, and the responses were typically supported by the retrieved evidence. Quantitative review showed that the chatbot allowed users to receive well structured responses that were both informative and led users to become more confident and satisfied with their answers. User feedback made note of the fact while praising its handling of subtle questions, suggesting further improvements to speed of response as well as processing of difficult questions in particular.

Overall, these results indicate that adding to RAG framework with LangChain's structured prompt generation and long context capabilities in LLMs subsume the quality of the responses to scientific and research based queries. Finally, this work demonstrates the potential of AI driven chat bots to address demanding queries, and establishes the foundations

towards advancement and more far reaching use in multi hop question answering in the technical domains.

V. CONCLUSION

Following this project, it can be observed that a Retrieval Auxiliary Augmented Generation (RAG) chatbot could answer a complex query related to research. The system is able to leverage the state-of-the-art tools such as Deepseek-r1 1.5B LLM, ChromaDB, LangChain etc and demonstrate the ability to process queries with extended context windows and generate nuanced, contextually relevant responses. Furthermore, actually running the system locally removed constraints stemming from online APIs like server busy errors, removing all practical limitations on total number of queries and embeddings.

With implementation, response accuracy improved, hallucinations were minimized, and high recall was achieved on technical and academic inquiries. It turned out to be an effective interpretation within scientific domains of progressive disease states, treatment modalities, and complicated contextual relationships. Similarly, further quantitative results and positive user feedback validated the chatbot's reliability in being as precise and coherent in its answers as possible. RAG frameworks and long context LLMs have great potential for developing advanced, evidence based conversational agents for particular fields, and this project illustrates that ability.

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